Lab Cleanup Robot using Mirte Master Platform

AUTHOR:

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Abstract

In this demo, we demonstrate how Jupyter Book can be used to create and publish a content rich paper that includes interactive elements such as code cells, visualizations, and multimedia. We will walk through the process of setting up a Jupyter Book, adding content, and deploying the final product online.

Keywords Navigation2, ROS2, SLAM, Mobile$_{Manipulation}$

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1 Introduction

Jupyter Book has been rebuild from ground up using the MyST engine (Jupyter et al., 2025). This allows to export content in multiple output formats including HTML, PDF and docx. In this paper we present an overview of the possibilities and demonstrate its working.

In an introduction. you often cite. Than can be done in various ways, either using a .bib file or directly using the doi.

cite with doi

- `[@doi:10.25080/hwcj9957]` resulting in (Jupyter et al., 2025)
- `@doi:10.25080/hwcj9957` resulting in Jupyter et al. (2025)

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- `{cite:t}`jupyter2025“ resulting in Jupyter et al. (2025)
- `{cite:p}`jupyter2025“ resulting in (Jupyter et al., 2025)

1.1 Background

Jupyter Book has been rebuild with the intend to export content in multiple output formats including HTML, PDF and docx. Figure 1 provides this idea.

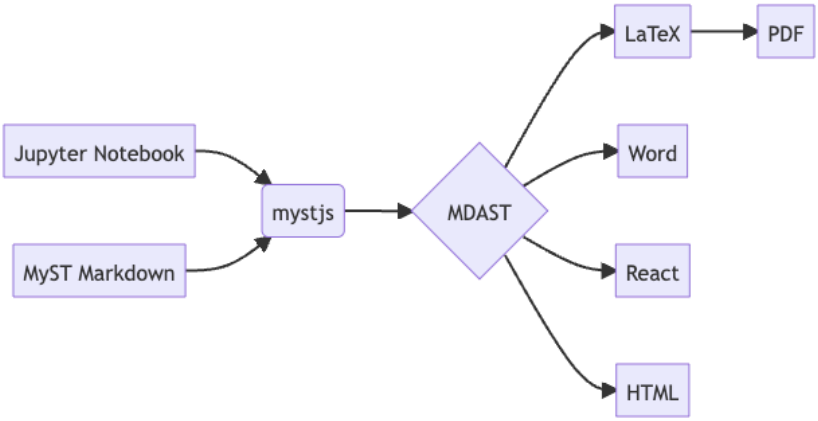

Figure 1: The myst engine allows Jupyter Notebook, markdown and even tex files to be converted to multiple output formats.

As exporting to different formats is possible, it is not always desired. Some content should only be visible in the HTML version, and some content only need to be included in the PDF version. You can use blocks like `+++{"no-pdf":true}` to enable this, as shown below where the figure is seen in the HTML version but not in the PDF version.

2 Theory

We can have inline equations like $E = mc^2$ or display equations like:

$$c\varphi \tag{1}$$

$$i\frac{\psi}{t} = -\frac{^2}{2m}{}^2\psi + V\psi \tag{2}$$

We can link to equations using their labels, like equation (2) or with more emphasis: (2). See the documentation for more options with using formulas. You might be interested in specific ways of numbering.

> **Tip**
> Check the hover-over functionality for these links!

3 Methodology

3.1 System Overview

The system essentially consists of a mobile manipulator (Mirte Master) tasked with autonomously exploring an indoor environment (the robotics lab), identifying objects, distinguishing between electronics and other objects, and sorting these objects accordingly.

the pipeline integrates three main components: **perception**, **coverage planning** and **high-level task strategy**.

3.1.1 Perception

Perception is responsible for object detection, classification and spatial localization. For this study we limited the object detection and localization approaches to two. A **2D detection and depth projection** approach, and 3D segmentation with **point cloud clustering**. As for classification methods, this study evaluates a classical cv approach, aswell as a classification model based approach.

objects are categorized into:

1. Graspable vs non-graspable
2. Electronics vs non-electronics

   iets meer over classification.

3.1.2 Coverage Strategy

Coverage planning refers to how the robot explores the mapped environment to look for graspable objects. Two coverage strategies are implemented:

- **Systematic coverage:** The environment is traversed using a structured pattern like a boustrophedon pattern to ensure complete coverage.
- **Reactive coverage**: The robot dynamically traverses and selects exploration targets based on frontier regions or detected objects, prioritizing areas of high information gain.

  iets meer over coverage planning.

3.1.3 System level strategy

The system level strategy determines when and how the objects are retrieved. Two high level behaviors are evaluated:

- Clean as you go: The robot attempts to grasp objects upon detection without mapping the environment fully first
- Map and optimize: The robot first explores and maps the environment and stores detected objects. Only after this stage does the robot collect objects based on a calculated plan.

  iets meer over system strategy.

3.2 Experimental design

This study follows factorial experimental design, analyzing the interplay between perception, coverage and system level decision making.

Three independent variables are defined:

- perception quality
  - Classical cv based classification

  - Model based classification

- detection method
  - point cloud clustering
  - depth projection

- coverage strategy
  - Systematic Coverage following a predefined path
  - Reactive Coverage by acting in real time

- System-level strategy
  - Clean-as-you-go
  - Map-then-optimize

this results in a total of 16 experimental configurations, however some of these setups are fundamentally incompatible, such as Systematic coverage with a clean as you go approach. The full setup is shown in table x.

3.3 Evaluation Metrics

3.3.1 Task level metrics

percentage of electronics, total completion time, failed attempts, distance traveled

3.3.2 Component level metrics

detection percision and recall, localization error, coverage percentage.

4 Results

With JupyterBook it is possible to make your narrative and data-analysis available in a single file, through Jupyter Notebooks. It is even possible to create interactive materials using widgets on the website, but leave them out in the PDF version (using tags). To run the code below, click the

icon in the top right corner. This will connect with a kernel. If the kernel is loaded, click

to run the code.

Figure 2 shows the results of fitting our linear model to the data. Parameter a is optimal when the chi-squared curve is at its minimum, which corresponds to the smallest residuals and the best match between model and data.

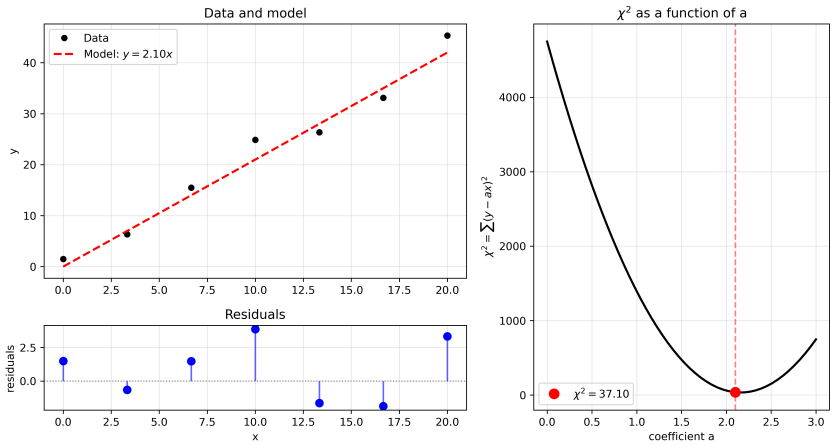

Figure 2: Fitting our data to a linear model with a single parameter a. The optimal value of a is found when the chi-squared curve (bottom) is at its minimum. The residuals (top right) are smallest at this point, and the model (top left) best matches the data.

5 Conclusion

In this study we investigated whether and how scientific publishing is feasible through the use of Jupyter Book. Using a starterkit template repository which can be easily accessed and used, a head start is provided. The manual provides detailed information to make more technical elements accessible. Hence, we conclude that Jupyter Book provides a feasible way of publishing scientific content - at least for bachelor and master thesis.

6 Appendix

6.1 Cheatsheet

6.2 README

This is the TU Delft starterkit for open publishing with Jupyter Book. It provides a skeleton with basic settings to have a head start with your thesis / project.

It includes:

- a github deploy file taking care of deploying the website and building a pdf
- basic content files that can be altered to fit your project
- yml files in the root folder to configure the book and the pdf build

6.2.1 Where to start

Already know about Jupyter Book, familiar with GitHub or GitLab, Markdown... start with cloning the starterkit and start writing working on your project. New to the ecosystem? Start with the manual, and then move on to the starterkit.

6.2.2 Purpose

The purpose of this project and this manual is to enable others to use Jupyter Book for writing and publishing their own scientific and educational content. We hope to lower the technical barrier for researchers and educators to create and share their own resources, and to promote open science and open education practices.

6.2.3 License

This book is licensed under the Creative Commons Attribution-NonCommercial 4.0 International License, unless stated otherwise.

You are free to:

- Share — copy and redistribute the material in any medium or format
- Adapt — remix, transform, and build upon the material for any non-commercial purpose provided proper attribution is given.

6.2.4 Errors

If you find any errors in the content, please report them by opening an issue on the GitHub repository.

6.2.5 Funding

This project has received funding from the Delft University of Technology Open Science Fund (2025-2026).

6.2.6 Identifier

TUDJBOSSTARTERKIT

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